

# Database Design

## Overview

This document serves as the single source of truth for the database design of the Animal Adoption Management System (AAMS). It covers all stages of the design process, from initial entity and relationship identification through to cardinality mapping, table definitions with column names and data types, and the final entity-relationship diagram. All design decisions are grounded in the functional requirements defined for the six user roles: Admin, Shelter Staff, Adopter, Veterinarian, Volunteer, and Donor.

## 1. Entities & Relationships

This section maps each functional requirement to its corresponding entities and relationships. Relationships are expressed in the format ENTITY\_A — verb — ENTITY\_B, where each entry represents a structural dependency that will translate into a table, foreign key, or junction table in the final schema. Query-only operations (such as search and filter) are excluded, as they do not produce new schema elements. Relationships are grouped by user role to reflect the access and ownership model of the system.

### 1.1 Adopter

| No | Relationship                             |
|----|------------------------------------------|
| 01 | ADOPTER - has - GOVERNMENT_ID            |
| 02 | ADOPTER - has - FAVORITES                |
| 03 | ADOPTER - submits - ADOPTION_APPLICATION |
| 04 | ADOPTER - receives - NOTIFICATIONS       |
| 05 | ADOPTER - cancels - VISITS               |

### 1.2 Shelter Staff

| No | Relationship                      |
|----|-----------------------------------|
| 01 | STAFF - has - GOVERNMENT_ID       |
| 02 | STAFF - updates - PET             |
| 03 | STAFF - belongs_to - SHELTER      |
| 04 | STAFF - manages - APPLICATIONS    |
| 05 | STAFF - updates - VOLUNTEERS      |
| 06 | STAFF - updates - VOLUNTEER_TASKS |

| No | Relationship                     |
|----|----------------------------------|
| 07 | STAFF - updates - SHELTER_EVENTS |
| 08 | STAFF - updates - DONATIONS      |
| 09 | STAFF - manages - VISITS         |

## 1.3 Pet

| No | Relationship                   |
|----|--------------------------------|
| 01 | PET - belongs_to - BREED       |
| 02 | BREED - belongs_to - SPECIES   |
| 03 | PET - has - HEALTH_RECORD      |
| 04 | PET - has - VACCINATION_RECORD |
| 05 | PET - has - TRANSFER_HISTORY   |

## 1.4 Veterinarian

| No | Relationship                       |
|----|------------------------------------|
| 01 | VET - has - GOVERNMENT_ID          |
| 02 | VET - manages - APPOINTMENTS       |
| 03 | VET - updates - VACCINATION_RECORD |
| 04 | VET - updates - HEALTH_RECORD      |

## 1.5 Volunteer

| No | Relationship                                |
|----|---------------------------------------------|
| 01 | VOLUNTEER - HAS - GOVERNMENT_ID             |
| 01 | VOLUNTEER - submits - VOLUNTEER_APPLICATION |
| 02 | VOLUNTEER - has - TASKS                     |
| 03 | VOLUNTEER - manages - SHELTER_EVENTS        |

## 1.6 DONOR

| No | Requirement               |
|----|---------------------------|
| 01 | DONOR - makes - DONATIONS |

## 1.7 Admin

| No | Requirement                 |
|----|-----------------------------|
| 01 | ADMIN - has - GOVERNMENT_ID |
| 02 | ADMIN - creates - SHELTER   |
| 03 | ADMIN - assigns - STAFF     |

## 2. Cardinality

This section defines the cardinality of each relationship identified in Section 1. Cardinality describes how many instances of one entity can be associated with instances of another, and is expressed as 1:1, 1:M, or M:M. Many-to-many relationships are resolved into junction tables, which are listed alongside the relationship. These mappings directly inform the placement of foreign keys and the creation of junction tables in the final schema.

| No | Relationship                             | Cardinality | Junction                              |
|----|------------------------------------------|-------------|---------------------------------------|
| 01 | ADOPTER - has - GOVERNMENT_ID            | 1:1         | -                                     |
| 02 | ADOPTER - has - FAVORITES                | M:M         | FAVORITES (adopterID, petID)          |
| 03 | ADOPTER - submits - ADOPTION_APPLICATION | 1:M         | -                                     |
| 04 | ADOPTER - cancels - VISITS               | 1:M         | -                                     |
| 05 | ADOPTER - receives - NOTIFICATIONS       | 1:M         | -                                     |
| 06 | STAFF - has - GOVERNMENT_ID              | 1:1         | -                                     |
| 07 | STAFF - updates - PET                    | 1:M         | -                                     |
| 08 | STAFF - belongs_to - SHELTER             | M:1         | -                                     |
| 09 | STAFF - manages - ADOPTION_APPLICATIONS  | 1:M         | -                                     |
| 10 | STAFF - updates - VOLUNTEER              | 1:M         | -                                     |
| 11 | STAFF - assigns - TASKS                  | 1:M         | -                                     |
| 12 | VOLUNTEER - handles - TASKS              | M:M         | VOLUNTEER_TASKS (taskID, volunteerID) |
| 13 | STAFF - updates - EVENTS                 | 1:M         | -                                     |
| 14 | STAFF - creates - DONOR                  | 1:M         | -                                     |
| 15 | STAFF - manages - VISITS                 | 1:M         | -                                     |
| 16 | PET - belongs_to - BREED                 | M:1         | -                                     |
| 17 | BREED - belongs_to - SPECIES             | M:1         | -                                     |
| 18 | PET - has - HEALTH_RECORD                | 1:M         | -                                     |
| 19 | PET - has - VACCINATION_RECORD           | 1:M         | -                                     |

| No | Relationship                                | Cardinality | Junction                                  |
|----|---------------------------------------------|-------------|-------------------------------------------|
| 20 | PET - has - TRANSFER_HISTORY                | 1:M         | -                                         |
| 21 | VET - manages - APPOINTMENTS                | 1:M         | -                                         |
| 22 | VET - updates - VACCINATION_RECORD          | 1:M         | -                                         |
| 23 | VET - updates - HEALTH_RECORD               | 1:M         | -                                         |
| 24 | VOLUNTEER - submits - VOLUNTEER_APPLICATION | 1:1         | -                                         |
| 25 | VOLUNTEER - has - TASKS                     | 1:M         | -                                         |
| 26 | VOLUNTEER - manages - SHELTER_EVENTS        | M:M         | VOLUNTEER_EVENT<br>(eventID, volunteerID) |
| 27 | DONOR - makes - DONATIONS                   | 1:M         | -                                         |
| 28 | ADMIN - creates - SHELTER                   | 1:M         | -                                         |
| 29 | ADMIN - assigns - STAFF                     | 1:M         | -                                         |
| 30 | APPOINTMENT - records - VACCINATION_RECORD  | 1:M         | -                                         |
| 31 | STAFF - assists - APPOINTMENTS              | 1:M         | -                                         |
| 32 | VOLUNTEER - assists - APPOINTMENTS          | 1:M         | -                                         |

## 3. Data Tables

This section defines the complete schema for each of the tables in the system. For every table, all columns are listed with their data type, constraints, and a brief description of the field's purpose. Primary keys and foreign keys are explicitly marked. These definitions serve as the direct input for the Prisma schema and database migrations.

### 3.1 USERS

Central authentication table shared by all six role tables (Admin, Staff, Veterinarian, Adopter, Volunteer, Donor) via userID as a shared primary/foreign key.

| Column Name         | Data Type        | Constraints           | Description                                                   |
|---------------------|------------------|-----------------------|---------------------------------------------------------------|
| <b>userID</b>       | INT              | PK,<br>AUTO_INCREMENT | Unique identifier for the user, shared across all role tables |
| <b>userEmail</b>    | VARCHAR<br>(45)  | NOT NULL, UNIQUE      | Login email address                                           |
| <b>userPassword</b> | VARCHAR<br>(255) | NOT NULL              | Bcrypt hashed password                                        |
| <b>refreshToken</b> | TEXT             | NULLABLE              | Hashed refresh token for the two-token auth flow              |

| Column Name          | Data Type    | Constraints                                                                          | Description                                                  |
|----------------------|--------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>role</b>          | VARCHAR (20) | NOT NULL<br>ENUM ('Adopter', 'Volunteer', 'Donor', 'Admin', 'Staff', 'Veterinarian') | Designates which role table this user's profile row lives in |
| <b>emailVerified</b> | BOOLEAN      | DEFAULT FALSE                                                                        | Email verification status                                    |
| <b>lastLoginAt</b>   | DATETIME     | NULLABLE                                                                             | Most recent successful login (all roles)                     |

## 3.2 ADMIN

Stores account and identity information for system administrators who manage shelter branches and staff assignments.

| Column Name               | Data Type     | Constraints                                                    | Description                                             |
|---------------------------|---------------|----------------------------------------------------------------|---------------------------------------------------------|
| <b>userID</b>             | INT           | PK, FK → USERS (userID)                                        | Unique identifier for the admin, shared with USERS      |
| <b>avatarSeed</b>         | VARCHAR (64)  | NOT NULL                                                       | Seed string used to generate the user's DiceBear avatar |
| <b>adminName</b>          | VARCHAR (45)  | NOT NULL                                                       | Name of the admin                                       |
| <b>createdAt</b>          | DATETIME      | NOT NULL                                                       | Account creation timestamp                              |
| <b>accountStatus</b>      | VARCHAR (45)  | DEFAULT 'Pending'<br>ENUM ('Pending', 'Active', 'Deactivated') | Account status                                          |
| <b>adminPhone</b>         | VARCHAR (20)  | NULLABLE                                                       | Admin contact number                                    |
| <b>addressLine1</b>       | VARCHAR (100) | DEFAULT ''                                                     | Street address, line 1                                  |
| <b>addressLine2</b>       | VARCHAR (100) | NULLABLE                                                       | Street address, line 2                                  |
| <b>city</b>               | VARCHAR (45)  | DEFAULT ''                                                     | City                                                    |
| <b>state</b>              | VARCHAR (45)  | DEFAULT ''                                                     | State                                                   |
| <b>zip</b>                | VARCHAR (10)  | DEFAULT ''                                                     | Postal code                                             |
| <b>country</b>            | VARCHAR (45)  | DEFAULT ''                                                     | Country                                                 |
| <b>onboardingComplete</b> | BOOLEAN       | DEFAULT FALSE                                                  | Whether the admin has finished the onboarding wizard    |

| Column Name               | Data Type | Constraints                                           | Description                                                               |
|---------------------------|-----------|-------------------------------------------------------|---------------------------------------------------------------------------|
| <b>onboardingStep</b>     | INT       | DEFAULT 2                                             | Furthest onboarding step reached (only ever advances)                     |
| <b>adminDOB</b>           | DATE      | NULLABLE                                              | Date of birth of admin                                                    |
| <b>adminSex</b>           | CHAR (1)  | NULLABLE                                              | Gender of admin                                                           |
| <b>statusChangedBy ID</b> | INT       | FK → ADMIN (userID)<br>NULLABLE<br>ON DELETE SET NULL | The admin who last changed this admin's accountStatus (self-referencing)  |
| <b>statusChangedAt</b>    | DATETIME  | NULLABLE                                              | Timestamp of the last accountStatus change, paired with statusChangedByID |

### 3.3 SHELTER

Stores details for each shelter branch in the multi-shelter network including location, capacity and status.

| Column Name            | Data Type         | Constraints                                       | Description                               |
|------------------------|-------------------|---------------------------------------------------|-------------------------------------------|
| <b>shelterID</b>       | INT               | PK,<br>AUTO_INCREMENT                             | Unique identifier for the shelter         |
| <b>shelterName</b>     | VARCHAR (45)      | NOT NULL                                          | Name of the shelter                       |
| <b>shelterAddress</b>  | VARCHAR (45)      | NOT NULL                                          | Shelter location                          |
| <b>shelterPhone</b>    | VARCHAR (20)      | NOT NULL                                          | Shelter phone number                      |
| <b>shelterEmail</b>    | VARCHAR (45)      | NOT NULL                                          | Shelter email address                     |
| <b>shelterLocation</b> | GEOGRAPHY (POINT) | NULLABLE                                          | Location of the shelter (PostGIS)         |
| <b>shelterZIP</b>      | INT               | NOT NULL                                          | Zip code of shelter                       |
| <b>shelterSize</b>     | INT               | NOT NULL                                          | Capacity of shelter                       |
| <b>shelterStatus</b>   | VARCHAR (10)      | DEFAULT 'Open'<br>ENUM ('Open', 'Full', 'Closed') | Shelter Status                            |
| <b>managerStaffID</b>  | INT               | FK → STAFF (userID)<br>NULLABLE                   | Unique identifier for the shelter manager |

### 3.4 STAFF

Stores account, contact and employment information for shelter staff members.

| Column Name               | Data Type     | Constraints                                                    | Description                                             |
|---------------------------|---------------|----------------------------------------------------------------|---------------------------------------------------------|
| <b>userID</b>             | INT           | PK, FK → USERS (userID)                                        | Unique identifier for the staff, shared with USERS      |
| <b>avatarSeed</b>         | VARCHAR (64)  | NOT NULL                                                       | Seed string used to generate the user's DiceBear avatar |
| <b>staffName</b>          | VARCHAR (45)  | NOT NULL                                                       | Full name of the staff                                  |
| <b>staffPhone</b>         | VARCHAR (20)  | NULLABLE                                                       | Staff contact number                                    |
| <b>addressLine1</b>       | VARCHAR (100) | DEFAULT ''                                                     | Street address, line 1                                  |
| <b>addressLine2</b>       | VARCHAR (100) | NULLABLE                                                       | Street address, line 2                                  |
| <b>city</b>               | VARCHAR (45)  | DEFAULT ''                                                     | City                                                    |
| <b>state</b>              | VARCHAR (45)  | DEFAULT ''                                                     | State                                                   |
| <b>zip</b>                | VARCHAR (10)  | DEFAULT ''                                                     | Postal code                                             |
| <b>country</b>            | VARCHAR (45)  | DEFAULT ''                                                     | Country                                                 |
| <b>onboardingComplete</b> | BOOLEAN       | DEFAULT FALSE                                                  | Whether the staff has finished the onboarding wizard    |
| <b>onboardingStep</b>     | INT           | DEFAULT 2                                                      | Furthest onboarding step reached (only ever advances)   |
| <b>shelterID</b>          | INT           | FK → SHELTER (shelterID)<br>NULLABLE                           | The shelter branch this staff works at                  |
| <b>staffDOB</b>           | DATE          | NULLABLE                                                       | Date of birth of staff                                  |
| <b>staffSex</b>           | CHAR (1)      | NULLABLE                                                       | Gender of staff                                         |
| <b>staffDOJ</b>           | DATETIME      | NULLABLE                                                       | Staff joining date                                      |
| <b>staffDOS</b>           | DATETIME      | NULLABLE                                                       | Staff date of separation                                |
| <b>staffDesignation</b>   | VARCHAR (45)  | NULLABLE<br>ENUM ('Manager', 'Senior', 'Associate')            | Staff designation                                       |
| <b>accountStatus</b>      | VARCHAR (45)  | DEFAULT 'Pending'<br>ENUM ('Pending', 'Active', 'Deactivated') | Account Status                                          |

## 3.5 VETERINARIAN

Stores account and contact information for vets who manage animal health appointments and records.

| Column Name               | Data Type     | Constraints                       | Description                                             |
|---------------------------|---------------|-----------------------------------|---------------------------------------------------------|
| <b>userID</b>             | INT           | PK, FK → USERS (userID)           | Unique identifier for the vet, shared with USERS        |
| <b>avatarSeed</b>         | VARCHAR (64)  | NOT NULL                          | Seed string used to generate the user's DiceBear avatar |
| <b>vetName</b>            | VARCHAR (45)  | NOT NULL                          | Name of the vet                                         |
| <b>addressLine1</b>       | VARCHAR (100) | DEFAULT ""                        | Street address, line 1                                  |
| <b>addressLine2</b>       | VARCHAR (100) | NULLABLE                          | Street address, line 2                                  |
| <b>city</b>               | VARCHAR (45)  | DEFAULT ""                        | City                                                    |
| <b>state</b>              | VARCHAR (45)  | DEFAULT ""                        | State                                                   |
| <b>zip</b>                | VARCHAR (10)  | DEFAULT ""                        | Postal code                                             |
| <b>country</b>            | VARCHAR (45)  | DEFAULT ""                        | Country                                                 |
| <b>onboardingComplete</b> | BOOLEAN       | DEFAULT FALSE                     | Whether the vet has finished the onboarding wizard      |
| <b>onboardingStep</b>     | INT           | DEFAULT 2                         | Furthest onboarding step reached (only ever advances)   |
| <b>vetPhone</b>           | VARCHAR (20)  | NULLABLE                          | Vet contact number                                      |
| <b>vetDOB</b>             | DATE          | NULLABLE                          | Date of Birth of vet                                    |
| <b>vetSex</b>             | CHAR (1)      | NULLABLE                          | Gender of vet                                           |
| <b>createdAt</b>          | DATETIME      | NOT NULL                          | Account creation timestamp                              |
| <b>shelterID</b>          | INT           | FK → SHELTER (shelterID) NULLABLE | Unique identifier for the shelter vet associated with   |

| Column Name          | Data Type    | Constraints                                                    | Description    |
|----------------------|--------------|----------------------------------------------------------------|----------------|
| <b>accountStatus</b> | VARCHAR (45) | DEFAULT 'Pending'<br>ENUM ('Pending', 'Active', 'Deactivated') | Account Status |

### 3.6 SPECIES

Stores the list of animal species supported by the system (e.g. Dog, Cat, Rabbit).

| Column Name        | Data Type    | Constraints           | Description                       |
|--------------------|--------------|-----------------------|-----------------------------------|
| <b>speciesID</b>   | INT          | PK,<br>AUTO_INCREMENT | Unique identifier for the species |
| <b>speciesName</b> | VARCHAR (45) | NOT NULL              | Name of the species               |

### 3.7 BREED

Stores animal breeds and their association to a parent species.

| Column Name      | Data Type    | Constraints              | Description                                            |
|------------------|--------------|--------------------------|--------------------------------------------------------|
| <b>breedID</b>   | INT          | PK,<br>AUTO_INCREMENT    | Unique identifier for the breed                        |
| <b>speciesID</b> | INT          | FK → SPECIES (speciesID) | Unique identifier for the species the breed belongs to |
| <b>breedName</b> | VARCHAR (45) | NOT NULL                 | Name of the breed                                      |

### 3.8 PET

Stores profile, intake, medical and compatibility information for all animals in the shelter network.

| Column Name    | Data Type    | Constraints                    | Description                                                                                     |
|----------------|--------------|--------------------------------|-------------------------------------------------------------------------------------------------|
| <b>petID</b>   | INT          | PK,<br>AUTO_INCREMENT          | Unique identifier for the pet.                                                                  |
| <b>petCode</b> | TEXT         | NULLABLE, UNIQUE,<br>GENERATED | Human-facing reference (e.g. "PE000002")<br>GENERATED ALWAYS<br>STORED from petID,<br>read-only |
| <b>petName</b> | VARCHAR (45) | NOT NULL                       | Name of the pet.                                                                                |
| <b>breedID</b> | INT          | FK → BREED (breedID)           | Pet Breed                                                                                       |
| <b>petDOB</b>  | DATE         | NOT NULL                       | Date of birth of the pet                                                                        |

| Column Name                   | Data Type     | Constraints                                                                                  | Description                                                                                    |
|-------------------------------|---------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>petWeight</b>              | FLOAT         | NOT NULL                                                                                     | Pet weight                                                                                     |
| <b>petHeight</b>              | FLOAT         | NOT NULL                                                                                     | Pet height                                                                                     |
| <b>petBGroup</b>              | VARCHAR (5)   | NOT NULL                                                                                     | Pet blood group                                                                                |
| <b>petColor</b>               | VARCHAR (45)  | NOT NULL                                                                                     | Pet color                                                                                      |
| <b>petSize</b>                | VARCHAR (10)  | NULLABLE<br>ENUM ('Small', 'Medium', 'Large')                                                | Size of pet                                                                                    |
| <b>petPhoto</b>               | VARCHAR (300) | NOT NULL                                                                                     | Primary profile photo URL for the Pet                                                          |
| <b>petSex</b>                 | CHAR (1)      | NOT NULL                                                                                     | Pet gender.                                                                                    |
| <b>petDesc</b>                | VARCHAR (500) | NULLABLE                                                                                     | Free-text description of the pet's personality, temperament and behaviour.                     |
| <b>microchipID</b>            | VARCHAR (45)  | NULLABLE                                                                                     | Industry-standard microchip identifier implanted on the pet. Nullable if not yet microchipped. |
| <b>intakeDate</b>             | DATE          | NOT NULL                                                                                     | Most recent intake date                                                                        |
| <b>intakeType</b>             | VARCHAR (45)  | NULLABLE<br>ENUM ('stray', 'surrendered', 'transferred')                                     | Most recent intake type                                                                        |
| <b>adoptionStatus</b>         | VARCHAR (45)  | NULLABLE<br>ENUM ('incoming', 'available', 'adopted', 'fostered', 'transferred', 'deceased') | Availability                                                                                   |
| <b>shelterID</b>              | INT           | FK → SHELTER (shelterID)                                                                     | The shelter branch where the pet is currently housed.                                          |
| <b>staffID</b>                | INT           | FK → STAFF (userID)<br>NULLABLE                                                              | Unique identifier for the staff member managing this pet's profile                             |
| <b>compatibleWithChildren</b> | BOOLEAN       | DEFAULT FALSE                                                                                | Compatibility with children                                                                    |
| <b>compatibleWithPets</b>     | BOOLEAN       | DEFAULT FALSE                                                                                | Compatibility with other pets                                                                  |

| Column Name         | Data Type | Constraints   | Description                                |
|---------------------|-----------|---------------|--------------------------------------------|
| <b>specialNeeds</b> | BOOLEAN   | DEFAULT FALSE | Special needs requirements                 |
| <b>featuredFlag</b> | BOOLEAN   | DEFAULT FALSE | Marks the pet as featured on the Home page |

### 3.9 PET\_PHOTO

Stores one or more photos per pet. Replaces the single petPhoto field with a one-to-many gallery; PET.petPhoto remains as the primary/profile photo URL, with additional photos stored here.

| Column Name       | Data Type     | Constraints               | Description                                         |
|-------------------|---------------|---------------------------|-----------------------------------------------------|
| <b>photoID</b>    | INT           | PK, AUTO_INCREMENT        | Unique identifier for the photo                     |
| <b>petID</b>      | INT           | FK → PET (petID)          | Unique identifier for the pet this photo belongs to |
| <b>photoURL</b>   | VARCHAR (300) | NOT NULL                  | Photo URL (Supabase Storage)                        |
| <b>uploadedAt</b> | DATETIME      | NOT NULL<br>DEFAULT now() | Upload timestamp                                    |

### 3.10 Adopter

Stores personal, household, lifestyle and account information for users who register to adopt or foster a pet.

| Column Name            | Data Type    | Constraints             | Description                                                |
|------------------------|--------------|-------------------------|------------------------------------------------------------|
| <b>userID</b>          | INT          | PK, FK → USERS (userID) | Unique identifier for the adopter, shared with USERS       |
| <b>avatarSeed</b>      | VARCHAR (64) | NOT NULL                | Seed string used to generate the user's DiceBear avatar    |
| <b>adopterName</b>     | VARCHAR (45) | NOT NULL                | Full name of the adopter                                   |
| <b>adopterDOB</b>      | DATE         | NULLABLE                | Date of birth of adopter                                   |
| <b>adopterSex</b>      | CHAR (1)     | NULLABLE                | Gender of adopter                                          |
| <b>createdAt</b>       | DATETIME     | NOT NULL                | Account creation timestamp                                 |
| <b>adopterRiskFlag</b> | BOOLEAN      | DEFAULT FALSE           | Flag set if AI feature detects suspicious adoption history |

| Column Name               | Data Type    | Constraints                                                             | Description                                                                                            |
|---------------------------|--------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>preQualifyFlag</b>     | BOOLEAN      | DEFAULT FALSE                                                           | Flag set if adopter is pre-qualified based on experience and history, used to fast-track applications. |
| <b>adopterPhone</b>       | VARCHAR (20) | NULLABLE                                                                | Adopter contact number                                                                                 |
| <b>housingType</b>        | VARCHAR (15) | NULLABLE<br>ENUM ('Apartment', 'House', 'Other')                        | House type                                                                                             |
| <b>ownsOrRents</b>        | VARCHAR (10) | NULLABLE<br>ENUM ('Owns', 'Rents')                                      | House ownership status                                                                                 |
| <b>landlordContact</b>    | VARCHAR (20) | NULLABLE                                                                | Landlord contact                                                                                       |
| <b>householdSize</b>      | INT          | NULLABLE                                                                | Household size                                                                                         |
| <b>numChildren</b>        | INT          | NULLABLE                                                                | Number of children at home                                                                             |
| <b>employmentStatus</b>   | VARCHAR (15) | NULLABLE<br>ENUM ('Unemployed', 'Student', 'Self_employed', 'Employed') | Employment status of the adopter                                                                       |
| <b>activityLevel</b>      | VARCHAR (10) | NULLABLE<br>ENUM ('Sedentary', 'Medium', 'Active')                      | Level of activity in lifestyle of adopter                                                              |
| <b>yardAvailable</b>      | BOOLEAN      | DEFAULT FALSE                                                           | Yard availability                                                                                      |
| <b>petExperience</b>      | VARCHAR (10) | NULLABLE<br>ENUM ('No', 'Little', 'Some', 'Very')                       | Amount of experience with pets                                                                         |
| <b>currentPets</b>        | INT          | DEFAULT 0                                                               | Number of pets currently owned by the adopter at time of registration.                                 |
| <b>preferredBreedID</b>   | INT          | FK → BREED (breedID)<br>NULLABLE                                        | Preferred breed for adoption                                                                           |
| <b>preferredAgeRange</b>  | VARCHAR (15) | NULLABLE<br>ENUM ('Young', 'Adult', 'Old')                              | Preferred pet age for adoption                                                                         |
| <b>preferredSize</b>      | VARCHAR (10) | NULLABLE<br>ENUM ('Small', 'Medium', 'Large')                           | Preferred pet size for adoption                                                                        |
| <b>openToSpecialNeeds</b> | BOOLEAN      | DEFAULT FALSE                                                           | Open to adopting special needs pets                                                                    |

| Column Name               | Data Type     | Constraints                                                  | Description                                            |
|---------------------------|---------------|--------------------------------------------------------------|--------------------------------------------------------|
| <b>adopterType</b>        | VARCHAR (10)  | NULLABLE<br>ENUM ('Fosterer', 'Owner')                       | Type of adopter                                        |
| <b>stripeCustomerID</b>   | VARCHAR (255) | NULLABLE                                                     | Stripe payment ID                                      |
| <b>accountStatus</b>      | VARCHAR (45)  | DEFAULT 'Active'<br>ENUM ('Active', 'Banned', 'Deactivated') | Account status                                         |
| <b>addressLine1</b>       | VARCHAR (100) | DEFAULT ''                                                   | Street address, line 1                                 |
| <b>addressLine2</b>       | VARCHAR (100) | NULLABLE                                                     | Street address, line 2                                 |
| <b>city</b>               | VARCHAR (45)  | DEFAULT ''                                                   | City                                                   |
| <b>state</b>              | VARCHAR (45)  | DEFAULT ''                                                   | State                                                  |
| <b>zip</b>                | VARCHAR (10)  | DEFAULT ''                                                   | Postal code                                            |
| <b>country</b>            | VARCHAR (45)  | DEFAULT ''                                                   | Country                                                |
| <b>onboardingComplete</b> | BOOLEAN       | DEFAULT FALSE                                                | Whether the adopter has finished the onboarding wizard |
| <b>onboardingStep</b>     | INT           | DEFAULT 2                                                    | Furthest onboarding step reached (only ever advances)  |

### 3.11 FAVORITES

Junction table linking adopters to pets they have saved to their favorites list.

| Column Name      | Data Type | Constraints               | Description                       |
|------------------|-----------|---------------------------|-----------------------------------|
| <b>adopterID</b> | INT       | PK, FK → ADOPTER (userID) | Unique identifier for the adopter |
| <b>petID</b>     | INT       | PK, FK → PET (petID)      | Unique identifier for the pet     |

### 3.12 ADOPTION\_APPLICATION

Stores adoption applications submitted by adopters for specific pets, including status and assigned staff.

| Column Name          | Data Type | Constraints           | Description                            |
|----------------------|-----------|-----------------------|----------------------------------------|
| <b>applicationID</b> | INT       | PK,<br>AUTO_INCREMENT | Unique identifier for the application. |
| <b>petID</b>         | INT       | FK → PET (petID)      | Unique identifier for the pet          |

| Column Name                    | Data Type     | Constraints                                                                | Description                                                                                                                |
|--------------------------------|---------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>adopterID</b>               | INT           | FK → ADOPTER (userID)                                                      | Unique identifier for the adopter                                                                                          |
| <b>createdAt</b>               | DATETIME      | NOT NULL                                                                   | Application creation timestamp                                                                                             |
| <b>staffID</b>                 | INT           | FK → STAFF (userID)<br>NULLABLE                                            | Staff ID of Staff handling the application                                                                                 |
| <b>shelterID</b>               | INT           | FK → SHELTER (shelterID)                                                   | Shelter ID of Shelter application is associated with.                                                                      |
| <b>applicationStatus</b>       | VARCHAR (20)  | DEFAULT 'Pending'<br>ENUM ('Pending', 'Accepted', 'Rejected', 'Withdrawn') | Application status                                                                                                         |
| <b>applicationCode</b>         | TEXT          | NULLABLE, UNIQUE, GENERATED                                                | Human-facing reference (e.g. "APP-00048") — Postgres GENERATED ALWAYS STORED column computed from applicationID, read-only |
| <b>applicationType</b>         | VARCHAR (10)  | DEFAULT 'Adopt',<br>ENUM ('Adopt', 'Foster')                               | Whether this is an adopt or foster application                                                                             |
| <b>shelterMessage</b>          | VARCHAR (500) | NULLABLE                                                                   | Adopter's message to the shelter, submitted with the application                                                           |
| <b>staffRemark</b>             | VARCHAR (300) | NULLABLE                                                                   | Staff-authored closing note shown to the adopter once resolved                                                             |
| <b>stripeCheckoutSessionID</b> | VARCHAR (255) | NOT NULL                                                                   | The Stripe Checkout Session that paid the application fee                                                                  |
| <b>stripePaymentIntentID</b>   | VARCHAR (255) | NULLABLE                                                                   | The associated Stripe PaymentIntent                                                                                        |
| <b>paymentStatus</b>           | VARCHAR (10)  | DEFAULT 'Paid',<br>ENUM ('Paid')                                           | Payment status of the processing fee                                                                                       |
| <b>amountPaid</b>              | DECIMAL (6,2) | DEFAULT 15.00                                                              | Application processing fee actually charged                                                                                |

### 3.13 VISITS

Stores shelter visit and virtual meet-and-greet appointments scheduled by adopters.

| Column Name      | Data Type | Constraints           | Description                       |
|------------------|-----------|-----------------------|-----------------------------------|
| <b>visitID</b>   | INT       | PK,<br>AUTO_INCREMENT | Unique identifier for the visit   |
| <b>adopterID</b> | INT       | FK → ADOPTER (userID) | Unique identifier for the adopter |

| Column Name        | Data Type     | Constraints                                              | Description                                                      |
|--------------------|---------------|----------------------------------------------------------|------------------------------------------------------------------|
| <b>petID</b>       | INT           | FK → PET (petID)<br>NULLABLE                             | Unique identifier for pet associated with the visit. Can be NULL |
| <b>staffID</b>     | INT           | FK → STAFF (userID)<br>NULLABLE                          | Unique identifier for Staff assisting the visit. Can be NULL     |
| <b>shelterID</b>   | INT           | FK → SHELTER (shelterID)                                 | Unique identifier for shelter associated with the visit.         |
| <b>visitTime</b>   | DATETIME      | NOT NULL                                                 | Visit date and time                                              |
| <b>remarks</b>     | VARCHAR (300) | NULLABLE                                                 | Any remarks related to the visit                                 |
| <b>visitStatus</b> | VARCHAR (45)  | NULLABLE<br>ENUM ('Confirmed', 'Cancelled', 'Completed') | Visit status — null until a staff member confirms it             |

### 3.14 NOTIFICATIONS

Stores system-generated automated reminders sent to users for vaccinations, appointments and tasks.

| Column Name               | Data Type     | Constraints                                            | Description                            |
|---------------------------|---------------|--------------------------------------------------------|----------------------------------------|
| <b>notificationID</b>     | INT           | PK,<br>AUTO_INCREMENT                                  | Unique identifier for the notification |
| <b>recipientID</b>        | INT           | NOT NULL                                               | Unique identifier for the recipient    |
| <b>notificationType</b>   | VARCHAR (20)  | ENUM ('vaccination', 'appointment', 'task')            | Type of notification                   |
| <b>notificationStatus</b> | VARCHAR (10)  | ENUM ('Pending', 'Sent', 'Failed')                     | Status of notification delivery        |
| <b>recipientType</b>      | VARCHAR (20)  | ENUM ('adopter', 'staff', 'volunteer', 'veterinarian') | Recipient user type                    |
| <b>sendTime</b>           | DATETIME      | NOT NULL                                               | Send date and time                     |
| <b>textBody</b>           | VARCHAR (300) | NOT NULL                                               | Notification text content              |

### 3.15 HEALTH\_RECORD

Stores medical history entries for each pet including diagnosis, treatment and vet notes.

| Column Name        | Data Type     | Constraints                            | Description                                                                                |
|--------------------|---------------|----------------------------------------|--------------------------------------------------------------------------------------------|
| <b>recordID</b>    | INT           | PK,<br>AUTO_INCREMENT                  | Unique identifier for the record                                                           |
| <b>petID</b>       | INT           | FK → PET (petID)                       | Unique identifier for the pet the record belongs to                                        |
| <b>createdAt</b>   | DATETIME      | NOT NULL                               | Creation date                                                                              |
| <b>lastUpdated</b> | DATETIME      | NOT NULL                               | Last update date                                                                           |
| <b>vetID</b>       | INT           | FK → VETERINARIAN (userID)<br>NULLABLE | Vet associated with record                                                                 |
| <b>recordDesc</b>  | VARCHAR (500) | NOT NULL                               | Description of the medical visit, diagnosis, treatment, or special needs noted by the vet. |

### 3.16 VACCINE

| Column Name         | Data Type     | Constraints           | Description                       |
|---------------------|---------------|-----------------------|-----------------------------------|
| <b>vaccineID</b>    | INT           | PK,<br>AUTO_INCREMENT | Unique identifier for the vaccine |
| <b>vaccineName</b>  | VARCHAR (45)  | NOT NULL              | Name of the vaccine               |
| <b>manufacturer</b> | VARCHAR (45)  |                       | Manufacturer name                 |
| <b>vaccineDesc</b>  | VARCHAR (500) |                       | Description of the vaccine        |

### 3.17 VACCINATION\_RECORD

Stores vaccination history for each pet including vaccine details, dates and administering vet.

| Column Name             | Data Type | Constraints              | Description                                         |
|-------------------------|-----------|--------------------------|-----------------------------------------------------|
| <b>recordID</b>         | INT       | PK,<br>AUTO_INCREMENT    | Unique identifier for the record                    |
| <b>petID</b>            | INT       | FK → PET (petID)         | Unique identifier for the pet the record belongs to |
| <b>vaccineID</b>        | INT       | FK → VACCINE (vaccineID) | Unique identifier for the vaccine administered      |
| <b>administeredDate</b> | DATETIME  | NOT NULL                 | Administered date                                   |
| <b>dueDate</b>          | DATETIME  | NOT NULL                 | Due date for administering vaccine                  |

| Column Name           | Data Type | Constraints                                                         | Description                                                                            |
|-----------------------|-----------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>administeredBy</b> | INT       | FK → VETERINARIAN (userID)<br>NULLABLE                              | Unique identifier for the vet administered the vaccine                                 |
| <b>administeredAt</b> | INT       | FK → SHELTER (shelterID)<br>NULLABLE                                | Unique identifier for the shelter where vaccine was administered                       |
| <b>appointmentID</b>  | INT       | FK → APPOINTMENTS (appointmentID)<br>NULLABLE<br>ON DELETE SET NULL | The appointment the dose was given at; null for doses recorded outside an appointment. |

### 3.18 TRANSFER\_HISTORY

Stores the history of inter-shelter pet transfers including the shelters and staff involved.

| Column Name             | Data Type     | Constraints                                                                         | Description                                                          |
|-------------------------|---------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>recordID</b>         | INT           | PK,<br>AUTO_INCREMENT                                                               | Unique identifier for the record                                     |
| <b>petID</b>            | INT           | FK → PET (petID)                                                                    | Unique identifier for the pet the record belongs to                  |
| <b>transferStatus</b>   | VARCHAR (20)  | DEFAULT 'In_Progress'<br>ENUM ('In_Progress', 'Completed', 'Rejected', 'Cancelled') | Transfer Status                                                      |
| <b>transferDate</b>     | DATETIME      | NOT NULL                                                                            | Date of transfer                                                     |
| <b>fromShelterID</b>    | INT           | FK → SHELTER (shelterID)                                                            | Unique identifier for the current shelter the pet is at              |
| <b>toShelterID</b>      | INT           | FK → SHELTER (shelterID)                                                            | Unique identifier for the new shelter the pet moving to              |
| <b>fromShelterStaff</b> | INT           | FK → STAFF (userID)<br>NULLABLE                                                     | Unique identifier for the staff handling transfer at current shelter |
| <b>toShelterStaff</b>   | INT           | FK → STAFF (userID)<br>NULLABLE                                                     | Unique identifier for the staff handling transfer at new shelter     |
| <b>transferReason</b>   | VARCHAR (300) | NOT NULL                                                                            | Reason for transfer                                                  |

### 3.19 APPOINTMENTS

Stores veterinary health appointments for pets including the attending vet and associated shelter.

| Column Name              | Data Type     | Constraints                                                         | Description                                                                                              |
|--------------------------|---------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>appointmentID</b>     | INT           | PK,<br>AUTO_INCREMENT                                               | Unique identifier for the appointmentID                                                                  |
| <b>appointmentCode</b>   | TEXT          | NULLABLE, UNIQUE,<br>GENERATED                                      | Human-facing reference (e.g. "APT-00123")<br>GENERATED ALWAYS<br>STORED from<br>appointmentID, read-only |
| <b>petID</b>             | INT           | FK → PET (petID)                                                    | Unique identifier for the pet associated with the appointment                                            |
| <b>vetID</b>             | INT           | FK → VETERINARIAN (userID)                                          | Unique identifier for the vet attending the appointment                                                  |
| <b>staffID</b>           | INT           | FK → STAFF (userID)<br>NULLABLE                                     | Staff member assigned                                                                                    |
| <b>volunteerID</b>       | INT           | FK → VOLUNTEER (userID)<br>NULLABLE                                 | Volunteer assigned                                                                                       |
| <b>shelterID</b>         | INT           | FK → SHELTER (shelterID)                                            | Unique identifier for the shelter the appointment is at                                                  |
| <b>appointmentDate</b>   | DATETIME      | NOT NULL                                                            | Appointment date and time                                                                                |
| <b>appointmentReason</b> | VARCHAR (300) | NOT NULL                                                            | Reason for appointment                                                                                   |
| <b>appointmentStatus</b> | VARCHAR (10)  | DEFAULT 'Scheduled'<br>ENUM ('Scheduled', 'Completed', 'Cancelled') | Appointment status                                                                                       |

### 3.20 VOLUNTEER\_APPLICATION

Stores applications submitted by prospective volunteers before their account is created.

| Column Name          | Data Type | Constraints                         | Description                                              |
|----------------------|-----------|-------------------------------------|----------------------------------------------------------|
| <b>applicationID</b> | INT       | PK,<br>AUTO_INCREMENT               | Unique identifier for the application                    |
| <b>volunteerID</b>   | INT       | FK → VOLUNTEER (userID)<br>NULLABLE | Volunteer ID created only if application is approved     |
| <b>staffID</b>       | INT       | FK → STAFF (userID)<br>NULLABLE     | Unique identifier for the staff handling the application |
| <b>createdAt</b>     | DATETIME  | NOT NULL                            | Application creation date and time.                      |
| <b>shelterID</b>     | INT       | FK → SHELTER (shelterID)            | Unique identifier for shelter                            |

| Column Name              | Data Type     | Constraints                                                   | Description              |
|--------------------------|---------------|---------------------------------------------------------------|--------------------------|
|                          |               |                                                               | volunteer is applying to |
| <b>applicationStatus</b> | VARCHAR (45)  | DEFAULT 'Created'<br>ENUM ('Created', 'Accepted', 'Rejected') | Application status       |
| <b>applicationRemark</b> | VARCHAR (100) | NULLABLE                                                      | Any remarks              |

### 3.21 VOLUNTEER

Stores account, contact and availability information for approved volunteers.

| Column Name               | Data Type     | Constraints                 | Description                                                                                    |
|---------------------------|---------------|-----------------------------|------------------------------------------------------------------------------------------------|
| <b>userID</b>             | INT           | PK, FK → USERS (userID)     | Unique identifier for the volunteer, shared with USERS                                         |
| <b>volunteerCode</b>      | TEXT          | NULLABLE, UNIQUE, GENERATED | Human-facing reference (e.g. "VOL-00005")<br>GENERATED ALWAYS<br>STORED from userID, read-only |
| <b>avatarSeed</b>         | VARCHAR (64)  | NOT NULL                    | Seed string used to generate the user's DiceBear avatar                                        |
| <b>volunteerName</b>      | VARCHAR (45)  | NOT NULL                    | Name of the volunteer                                                                          |
| <b>addressLine1</b>       | VARCHAR (100) | DEFAULT ""                  | Street address, line 1                                                                         |
| <b>addressLine2</b>       | VARCHAR (100) | NULLABLE                    | Street address, line 2                                                                         |
| <b>city</b>               | VARCHAR (45)  | DEFAULT ""                  | City                                                                                           |
| <b>state</b>              | VARCHAR (45)  | DEFAULT ""                  | State                                                                                          |
| <b>zip</b>                | VARCHAR (10)  | DEFAULT ""                  | Postal code                                                                                    |
| <b>country</b>            | VARCHAR (45)  | DEFAULT ""                  | Country                                                                                        |
| <b>onboardingComplete</b> | BOOLEAN       | DEFAULT FALSE               | Whether the volunteer has finished the onboarding wizard                                       |
| <b>onboardingStep</b>     | INT           | DEFAULT 2                   | Furthest onboarding step                                                                       |

| Column Name              | Data Type     | Constraints                                                              | Description                                            |
|--------------------------|---------------|--------------------------------------------------------------------------|--------------------------------------------------------|
|                          |               |                                                                          | reached (only ever advances)                           |
| <b>volunteerPhone</b>    | VARCHAR (20)  | NULLABLE                                                                 | Volunteer contact number                               |
| <b>volunteerDOB</b>      | DATE          | NULLABLE                                                                 | Date of Birth of volunteer                             |
| <b>volunteerSex</b>      | CHAR (1)      | NULLABLE                                                                 | Gender of volunteer                                    |
| <b>volunteerSchedule</b> | VARCHAR (100) | NULLABLE                                                                 | Schedule of volunteer                                  |
| <b>shelterID</b>         | INT           | FK → SHELTER (shelterID)<br>NULLABLE                                     | Unique identifier for shelter volunteer works at       |
| <b>createdAt</b>         | DATETIME      | NOT NULL                                                                 | Account creation timestamp                             |
| <b>accountStatus</b>     | VARCHAR (45)  | DEFAULT 'Pending'<br>ENUM ('Pending', 'Active', 'Banned', 'Deactivated') | Account status — Pending until a staff member approves |

### 3.22 GovernmentID

Stores government identity documents for adopters, staff, volunteers, vets and admins for verification purposes.

| Column Name               | Data Type     | Constraints                                                                 | Description                                                    |
|---------------------------|---------------|-----------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>governmentIDID</b>     | INT           | PK,<br>AUTO_INCREMENT                                                       | Unique identifier for the record.                              |
| <b>userID</b>             | INT           | NOT NULL                                                                    | The ID of the adopter, staff, volunteer, veterinarian or admin |
| <b>userType</b>           | VARCHAR (45)  | NOT NULL<br>ENUM ('Adopter', 'Staff', 'Volunteer', 'Veterinarian', 'Admin') | Designation                                                    |
| <b>idType</b>             | VARCHAR (45)  | NOT NULL                                                                    | Type of ID document                                            |
| <b>idNumber</b>           | VARCHAR (45)  | NOT NULL                                                                    | ID value                                                       |
| <b>verificationStatus</b> | VARCHAR (15)  | DEFAULT 'Pending'<br>ENUM ('Pending', 'Verified', 'Rejected')               | Verification status                                            |
| <b>documentURL</b>        | VARCHAR (500) | NULLABLE                                                                    | Object path within the private government-ids Storage bucket   |

### 3.23 TASKS

Stores volunteer tasks created by staff for a specific shelter branch.

| Column Name       | Data Type     | Constraints                                                                                           | Description                                              |
|-------------------|---------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>taskID</b>     | INT           | PK,<br>AUTO_INCREMENT                                                                                 | Unique identifier for the task                           |
| <b>taskName</b>   | VARCHAR (100) | NOT NULL<br>ENUM ('Animal_Care', 'Vet_Assistance', 'Cleaning', 'Feeding', 'Events', 'Admin', 'Other') | Task Name                                                |
| <b>shelterID</b>  | INT           | FK → SHELTER (shelterID)                                                                              | Unique identifier for the shelter where task is assigned |
| <b>taskDesc</b>   | VARCHAR (300) | NOT NULL                                                                                              | Task Description                                         |
| <b>taskDate</b>   | DATETIME      | NULLABLE                                                                                              | Task creation date                                       |
| <b>taskDue</b>    | DATETIME      | NULLABLE                                                                                              | Task due date                                            |
| <b>taskStatus</b> | VARCHAR (45)  | NULLABLE<br>ENUM ('In_progress', 'Completed', 'Cancelled')                                            | Task status                                              |
| <b>staffID</b>    | INT           | FK → STAFF (userID)<br>NULLABLE                                                                       | Unique identifier for the staff responsible for task     |

### 3.24 VOLUNTEER\_TASK

Junction table linking volunteers to the tasks assigned to them.

| Column Name        | Data Type | Constraints                 | Description                                          |
|--------------------|-----------|-----------------------------|------------------------------------------------------|
| <b>taskID</b>      | INT       | PK, FK → TASKS (taskID)     | Unique identifier for the task                       |
| <b>volunteerID</b> | INT       | PK, FK → VOLUNTEER (userID) | Unique identifier for the volunteer assigned to task |

### 3.25 EVENTS

Stores adoption events, fundraisers and community programs organised by shelter staff.

| Column Name      | Data Type | Constraints              | Description                                                |
|------------------|-----------|--------------------------|------------------------------------------------------------|
| <b>eventID</b>   | INT       | PK,<br>AUTO_INCREMENT    | Unique identifier for the event                            |
| <b>shelterID</b> | INT       | FK → SHELTER (shelterID) | Unique identifier for the shelter where event takes place. |

| Column Name          | Data Type     | Constraints                                                                                                                                                                     | Description                                                    |
|----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>eventName</b>     | VARCHAR (45)  | NOT NULL                                                                                                                                                                        | Name of event                                                  |
| <b>eventCategory</b> | VARCHAR (25)  | NOT NULL<br>ENUM ('Adoption_Event',<br>'Fundraiser',<br>'Volunteer_Orientation',<br>'Vaccination_Clinic',<br>'Community_Outreach',<br>'Workshop',<br>'Donation_Drive', 'Other') | Event category                                                 |
| <b>eventDate</b>     | DATETIME      | NOT NULL                                                                                                                                                                        | Event date and time                                            |
| <b>eventDesc</b>     | VARCHAR (300) | NOT NULL                                                                                                                                                                        | Description of event.                                          |
| <b>staffID</b>       | INT           | FK → STAFF (userID)<br>NULLABLE                                                                                                                                                 | Unique identifier for the staff responsible for managing event |

### 3.26 VOLUNTEER\_EVENT

Junction table linking volunteers to the events they are participating in.

| Column Name        | Data Type | Constraints                 | Description                                           |
|--------------------|-----------|-----------------------------|-------------------------------------------------------|
| <b>eventID</b>     | INT       | PK, FK → EVENTS (eventID)   | Unique identifier for the event                       |
| <b>volunteerID</b> | INT       | PK, FK → VOLUNTEER (userID) | Unique identifier for the volunteer assigned to event |

### 3.27 DONOR

Stores account and contact information for users who make monetary donations to shelter branches.

| Column Name         | Data Type     | Constraints             | Description                                             |
|---------------------|---------------|-------------------------|---------------------------------------------------------|
| <b>userID</b>       | INT           | PK, FK → USERS (userID) | Unique identifier for the donor, shared with USERS      |
| <b>avatarSeed</b>   | VARCHAR (64)  | NOT NULL                | Seed string used to generate the user's DiceBear avatar |
| <b>donorName</b>    | VARCHAR (45)  | NOT NULL                | Name of donor                                           |
| <b>addressLine1</b> | VARCHAR (100) | DEFAULT ""              | Street address, line 1                                  |
| <b>addressLine2</b> | VARCHAR (100) | NULLABLE                | Street address, line 2                                  |
| <b>city</b>         | VARCHAR (45)  | DEFAULT ""              | City                                                    |

| Column Name               | Data Type     | Constraints                                                  | Description                                           |
|---------------------------|---------------|--------------------------------------------------------------|-------------------------------------------------------|
|                           |               |                                                              |                                                       |
| <b>state</b>              | VARCHAR (45)  | DEFAULT ''                                                   | State                                                 |
| <b>zip</b>                | VARCHAR (10)  | DEFAULT ''                                                   | Postal code                                           |
| <b>country</b>            | VARCHAR (45)  | DEFAULT ''                                                   | Country                                               |
| <b>onboardingComplete</b> | BOOLEAN       | DEFAULT FALSE                                                | Whether the donor has finished the onboarding wizard  |
| <b>onboardingStep</b>     | INT           | DEFAULT 2                                                    | Furthest onboarding step reached (only ever advances) |
| <b>donorPhone</b>         | VARCHAR (20)  | NULLABLE                                                     | Donor contact number                                  |
| <b>donorDOB</b>           | DATE          | NULLABLE                                                     | Date of Birth of donor                                |
| <b>donorSex</b>           | CHAR (1)      | NULLABLE                                                     | Gender of donor                                       |
| <b>createdAt</b>          | DATETIME      | NOT NULL                                                     | Account creation timestamp                            |
| <b>stripeCustomerID</b>   | VARCHAR (255) | NULLABLE                                                     | Stripe payment ID                                     |
| <b>accountStatus</b>      | VARCHAR (45)  | DEFAULT 'Active'<br>ENUM ('Active', 'Banned', 'Deactivated') | Account status                                        |

### 3.28 DONATION

Stores individual donation transactions made by donors to specific shelter branches.

| Column Name         | Data Type | Constraints                    | Description                                                                                  |
|---------------------|-----------|--------------------------------|----------------------------------------------------------------------------------------------|
| <b>donationID</b>   | INT       | PK,<br>AUTO_INCREMENT          | Unique identifier for the donation                                                           |
| <b>donationCode</b> | TEXT      | NULLABLE, UNIQUE,<br>GENERATED | Human-facing reference (e.g. "DON-00012") GENERATED ALWAYS STORED from donationID, read-only |
| <b>donorID</b>      | INT       | FK → DONOR (userID)            | Unique identifier for the donor                                                              |
| <b>shelterID</b>    | INT       | FK → SHELTER (shelterID)       | Unique identifier for the shelter                                                            |
| <b>donationDate</b> | DATETIME  | NOT NULL<br>DEFAULT now()      | Donation date and time                                                                       |

| Column Name         | Data Type     | Constraints | Description             |
|---------------------|---------------|-------------|-------------------------|
| <b>donationAmt</b>  | FLOAT         | NOT NULL    | Amount donated.         |
| <b>donationDesc</b> | VARCHAR (300) | NULLABLE    | Description of donation |

## 3.29 Constraints & Indexes

Lists the database-level constraints and indexes that sit outside individual column definitions, including partial, expression and spatial indexes.

| Table                       | Constraint                                                                   | Purpose                                                        |
|-----------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>ADOPTION_APPLICATION</b> | UNIQUE (adopterID, petID) WHERE applicationStatus IN ('Pending', 'Accepted') | One active application per adopter + pet                       |
| <b>ADOPTION_APPLICATION</b> | UNIQUE (petID) WHERE applicationStatus = 'Accepted'                          | One accepted application per pet — a pet goes to one applicant |
| <b>APPOINTMENTS</b>         | UNIQUE (petID, vetID, appointmentDate) WHERE appointmentStatus = 'Scheduled' | Blocks double-submitted bookings                               |
| <b>GOVERNMENT_ID</b>        | UNIQUE (userID, userType)                                                    | One ID per user                                                |
| <b>SPECIES</b>              | UNIQUE (LOWER(speciesName))                                                  | Case-insensitive unique names                                  |
| <b>BREED</b>                | UNIQUE (speciesID, LOWER(breedName))                                         | Case-insensitive unique within a species                       |
| <b>SHELTER</b>              | GIST index on shelterLocation                                                | Fast PostGIS proximity search                                  |

Note: Partial/expression indexes and generated columns can't be expressed in the Prisma schema, so they live in migration SQL (the baseline migration builds the full schema).

## 4. ER-Diagram

ER-Diagram: <https://dbdiagram.io/d/PetPals-69bf5ca6fb2db18e3bd4b303>